

Lecture 01

Course Introduction & Review on Basic Probability Concepts

Course Overview

❖ Statistical Hydrology (통계수문학)

- ◆ 강의 목표: 불확실한 수문 변수를 적절한 방법으로 분석할 수 있고, 극한 사상에 대한 확률 분석을 수행할 수 있으며, 여러 가지의 회귀분석을 대상 자료를 사용하여 시행할 수 있다.
 - 여러 가지 분포의 특성을 이해하고, 실제 자료를 분포에 적합할 수 있다.
 - 극치값들에 대한 분석을 극치분포나 POT모델을 사용하여 분석할 수 있다.
 - 트렌드 분석, 회귀분석, 시계열 분석 등의 개념을 익히고 사용할 수 있다.
- * 모든 적용 및 분석은 컴퓨터 언어 (R) 를 통해 적용, **NOT** Microsoft Excel

◆ 강의 키워드 (Keywords of the course)

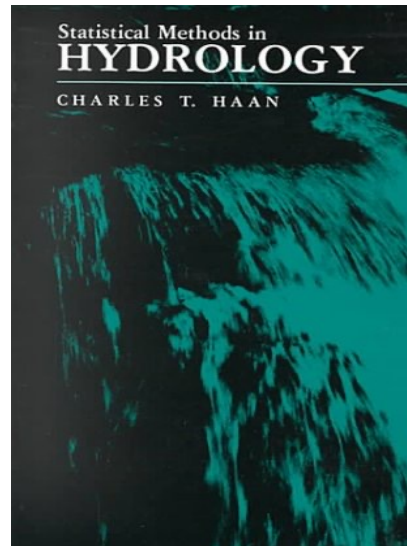
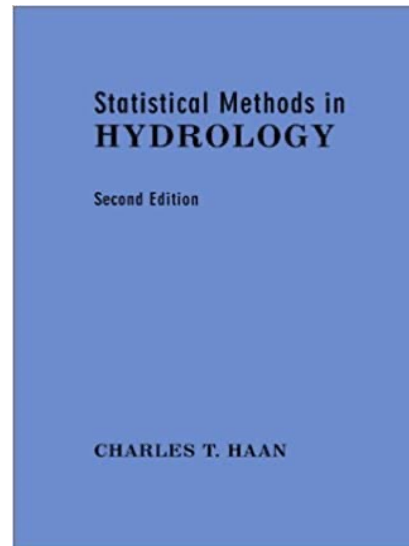
- Part I (Weeks 1-5): Probabilistic Modeling in Hydrology
- Part II (Weeks 6-9): Extreme Event Analysis

Statistical Hydrology, Part III (Weeks 10-14): Trend Detection, Regression Models, Time Series Analysis

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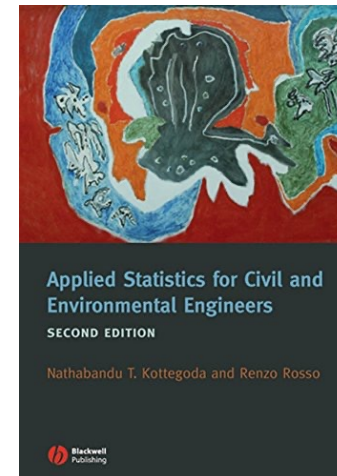
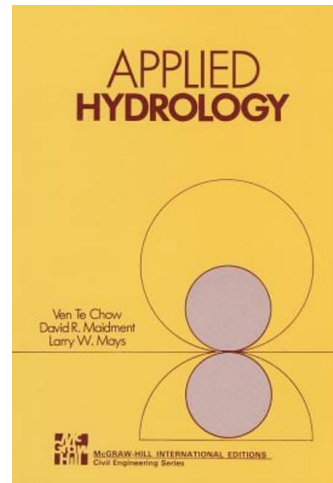
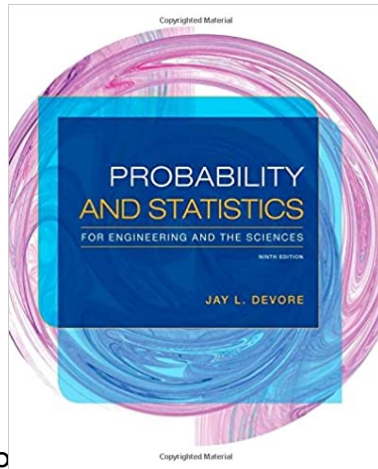
Primary Textbook

- ❖ Hann, C. T. (2002). **Statistical Methods in Hydrology (Second Edition)**. Iowa State Press.
 - ◆ Both Lecture Notes & In-class Assignments
 - ◆ PDF Available upon request



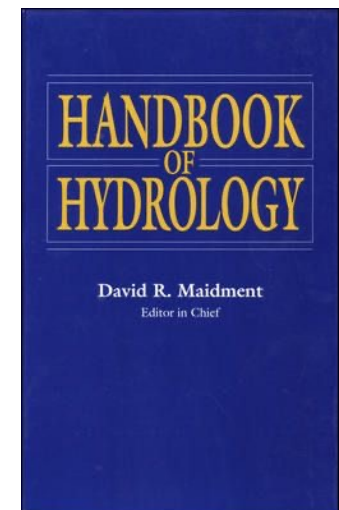
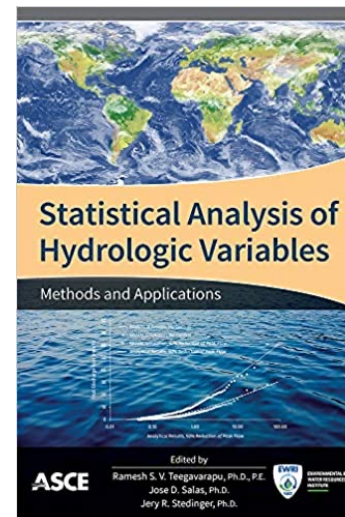
Supplementary Textbooks

- ❖ Devore, J. L. (2016). **Probability and Statistics for Engineering and the Sciences, Ninth Edition**. Cengage Learning.
- ❖ Chow, V. T. & Maidment, D. R. (1988). **Applied Hydrology**. McGraw-Hill, Inc. (Especially Chapter 11 Hydrologic Statistics & Chapter 12 Frequency Analysis)
- ❖ Kottegoda, N. T. & Rosso, R. (1997). **Statistics, Probability, and Reliability for Civil and Environmental Engineers**. McGraw-Hill.



Supplementary Textbooks

- ❖ Teegavarapu, R. S. V., Salas, J. D., & Stedinger, J. R. (2019). **Statistical Analysis of Hydrologic Variables**. ASCE.
- ❖ Maidment, D. R. (1993). **Handbook of Hydrology**. McGraw-Hill, Inc. (Especially Chapters 17-20:
 - Chapter 17 Statistical Treatment of Hydrologic Data
 - Chapter 18 Frequency Analysis of Extreme Events
 - Chapter 19 Analysis and Modeling of Hydrologic Time Series
 - Chapter 20 Geostatistics)



Course Schedule (Fall 2022)

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소

COURSE SCHEDULE (FALL 2022)

WEEK/DATE	TOPIC	ASSIGNMENTS & QUIZZES
1 Sep-06	<i>Course Introduction</i> Lec 01: Review on Basic Probability Concepts	
2 Sep-13	Lec 02: Probabilistic Modeling in Hydrology- Part I (Moments & Probabilistic Models)	
3 Sep-20	Lec 03: Probabilistic Modeling in Hydrology- Part II (MLE & Goodness of fit tests)	Quiz 1
4 Sep-27	<i>Training with Probabilistic Modeling</i>	
5 Oct-04	Field Trip (Details TBD)	
6 Oct-11	Lec 04: Extreme Event Analysis- Part I (GEV Distribution & Uncertainties)	Assignment 1 Due
7 Oct-18	Lec 05: Extreme Event Analysis- Part II (LP3 Distributions & POT Models)	
8 Oct-25	<i>Term Project Presentation: Bulletin 17B/17C</i>	



VWORLD 공간정보 오픈플랫폼 | 환경부

한강홍수통제소 (예정),
자세한 사항은 추후 공지 예정

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	Lec 06: Extreme Event Analysis- Part III (Selecting thresholds & Estimation using POT models)	Quiz 2
	<i>Training with Extreme Event Analysis</i> Lec 07: Introduction to Trend Detection	
	Lec 08: Analysis using Regression Models <i>Training with Regression</i>	
12 Nov-22	TEST (Closed Book)	Assignment 2 Due
13 Nov-29	Lec 09: Introduction to Time Series Analysis	
14 Dec-06	Lec 10: Hydrologic Statistics Frontiers -Introduction and Examples	
15 Dec-13	<i>Final Project Presentation</i>	Final Project Report Due Dec-20

Field Trip (WRSE, Spring 2022)

- ❖ 보령 다목적댐 & 정수장
- ❖ 보령댐-백제보 도수로



Introduction to the Course

❖ STATISTICAL HYDROLOGY (통계수문학)

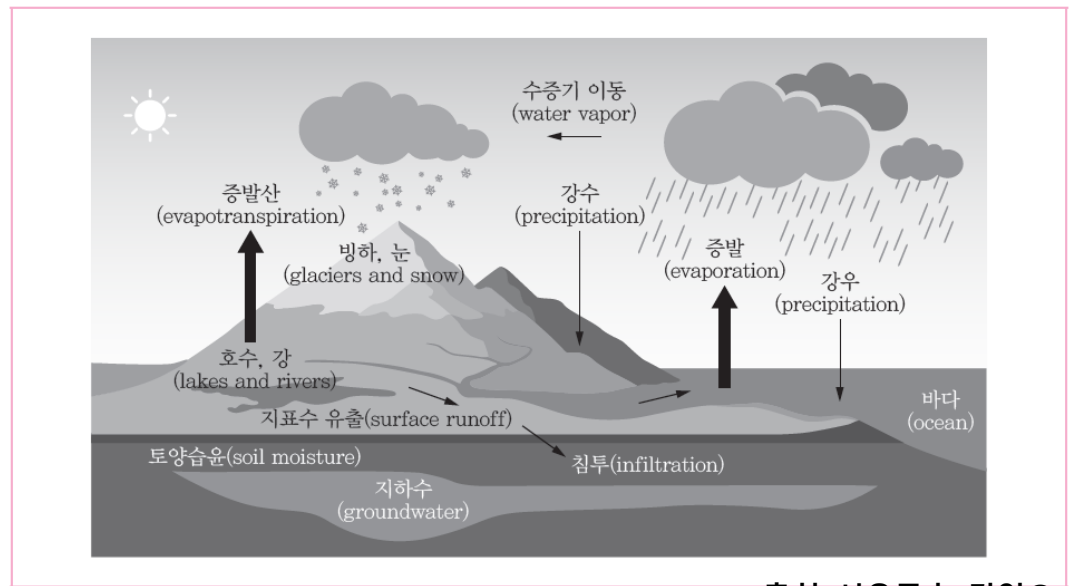
1) 물의 근원, 분배, 소멸 과정을 연구하는 것으로 정의될 수 있는 수문학

- 예를 들어, 물의 발생 원리, 강우-유출 관계, 지하수의 움직임 등을 포함 (수리학: 흐르는 또는 정지된 물에 구조물이 있을 경우 이들과의 관계를 연구하는 학문)

◆ 수문학의 접근방법

- 모델링
- 최적화
- 실험 (관측)

2) 통계



| 그림 1.1 | 물의 순환(<http://stem.guide/the-hydrologic-cycle>에서 수정)

출처: 선우중호, 김영오
제 3판 수문학 (2019)

Introduction to the Course

집중호우...피해 속출

100년 빈도 폭우 내리는데...하수도 배수량은 10년 빈도

입력 2022.08.10 (14:11) | 수정 2022.08.10 (14:54)

취재K

11 4



9일 서울 강남구 대치사거리의 배수구가 역류한 뒤, 두경이 유실된 모습.

수도권을 중심으로 중부지방에 내린 집중호우는 큰 피해를 남겼습니다. 특히 건물이 밀집한 서울 강남에서 인명피해와 재산 피해가 집중됐습니다.

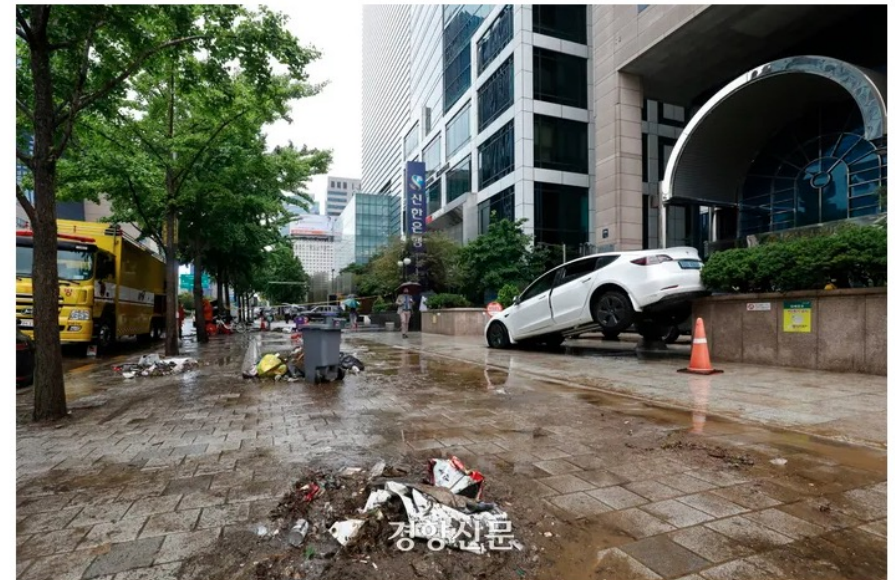
빗물이 빠져야 할 하수도가 도리어 역류하면서 도로가 물에 잠겼고 운행하던 차들은 침수된 도로 위를 강처럼 흘러내리면서 저지대 주거지와 지하상가가 침수돼 인명피해도 발생했습니다.

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가

'200년 빈도' 폭우가 일상, 도시 치수·방재 대책

입력 : 2022.08.09 15:46 | 수정 : 2022.08.09 23:49 김보미 기자 김원진 기자



서울 등 중부지역에 기록적인 폭우가 이어지고 있는 9일 침수 피해를 입은 서울 서초구 진흥아파트 앞 서초대로에 전날 쏟아진 폭우로 고립된 차량이 인도에 올라와 있다./문재원 기자

7

Introduction to the Course

- ❖ 수문학에서 불확실성은 아래와 같이 세 가지로 분류가 가능:
(Handbook of Hydrology, 17.1.1)
 - ◆ **Inherent randomness** of the driving variables (predominantly **precipitation**) and the randomness of the hydrologic system (including topographic, aquifer, and soil characteristics)
 - ◆ **Sampling error** – the measurements that hydrologists have to work with are only a small sample from a large or potentially infinite population
 - ◆ Uncertainty resulting from incorrect understanding of the processes involved (**System error**- errors in estimating or predicting system outputs from system inputs)
- ❖ How do we deal with uncertain variables in hydrology?
 - ◆ Replacing with **Representative Values (대표값) (Summary Statistics)**

Review on the Representative Values

❖ Basic Probability Concepts

1) Measures of Central Tendency

- Arithmetic Mean (expected values)
- Median (중앙값)
 - => useful for describing central tendency regardless of skewness (less sensitive to outsiders)
- Mode (최빈값)

$$\bar{X} = \sum_{i=1}^n \frac{x_i}{n}$$

$$\hat{X}_{0.5}$$

1, 3, 3, 6, 7, 8, 9

Median = **6**

1, 2, 3, 4, 5, 6, 8, 9

Median = (4+5) / 2 = **4.5**

(Where n is the number of data points)

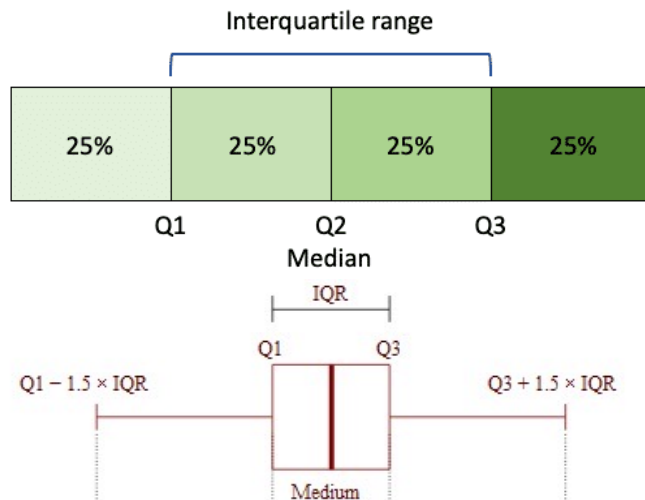
Review on the Representative Values

❖ Basic Probability Concepts

2) Measures of Dispersion: Describes the spread of data

◆ Examples include:

- Range: X_{max} & X_{min} (관측값중 가장 큰 값 & 가장 작은 값)
- Quartiles & Interquartile Range
- Variance (Standard Deviation)



The diagram shows a normal distribution curve with three overlapping curves labeled A (purple), B (pink), and C (blue). A vertical line marks the mean. Below the curve, the formula for Sample Variance is given:

$$S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

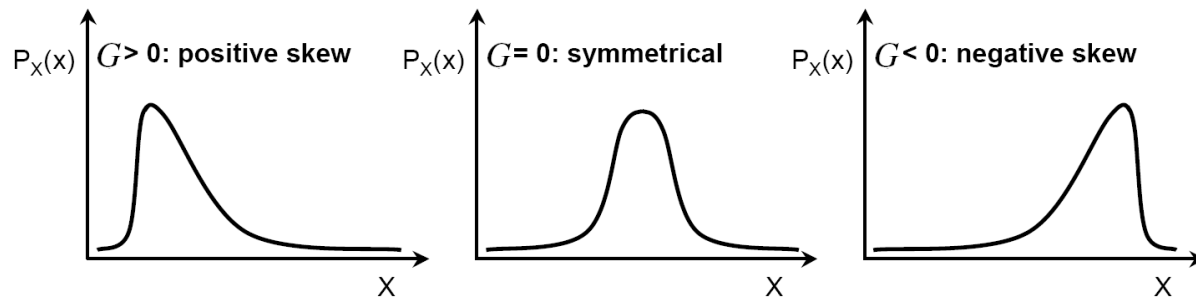
Review on the Representative Values

❖ Basic Probability Concepts

3) Measure of Asymmetry

- Coefficient of Skew

$$G = \frac{\tilde{M}_3}{S^3}$$



Using the 3rd moment and the standard deviation,

$$\tilde{M}_k \equiv \sum_{i=1}^n \frac{(X_i - \bar{X})^k}{n} \quad \text{and} \quad S \equiv \sqrt{S^2}$$

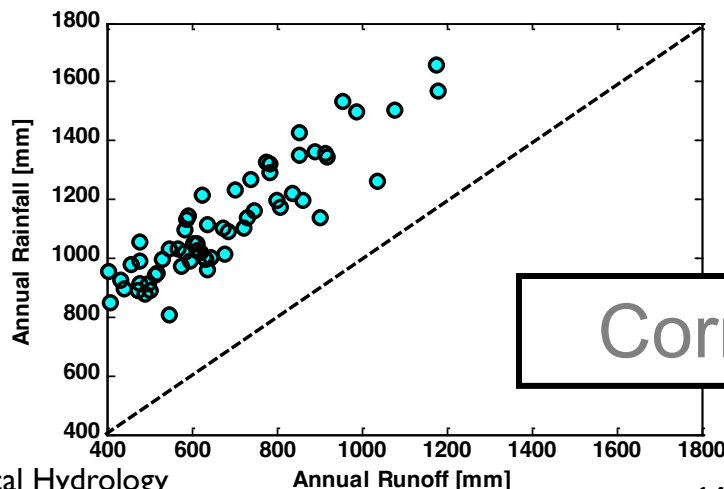
Review on the Representative Values

❖ Basic Probability Concepts (Data observed in Pairs)

4) Measure of Correlation (Relation between two random variables)

- Linear Correlation Coefficient

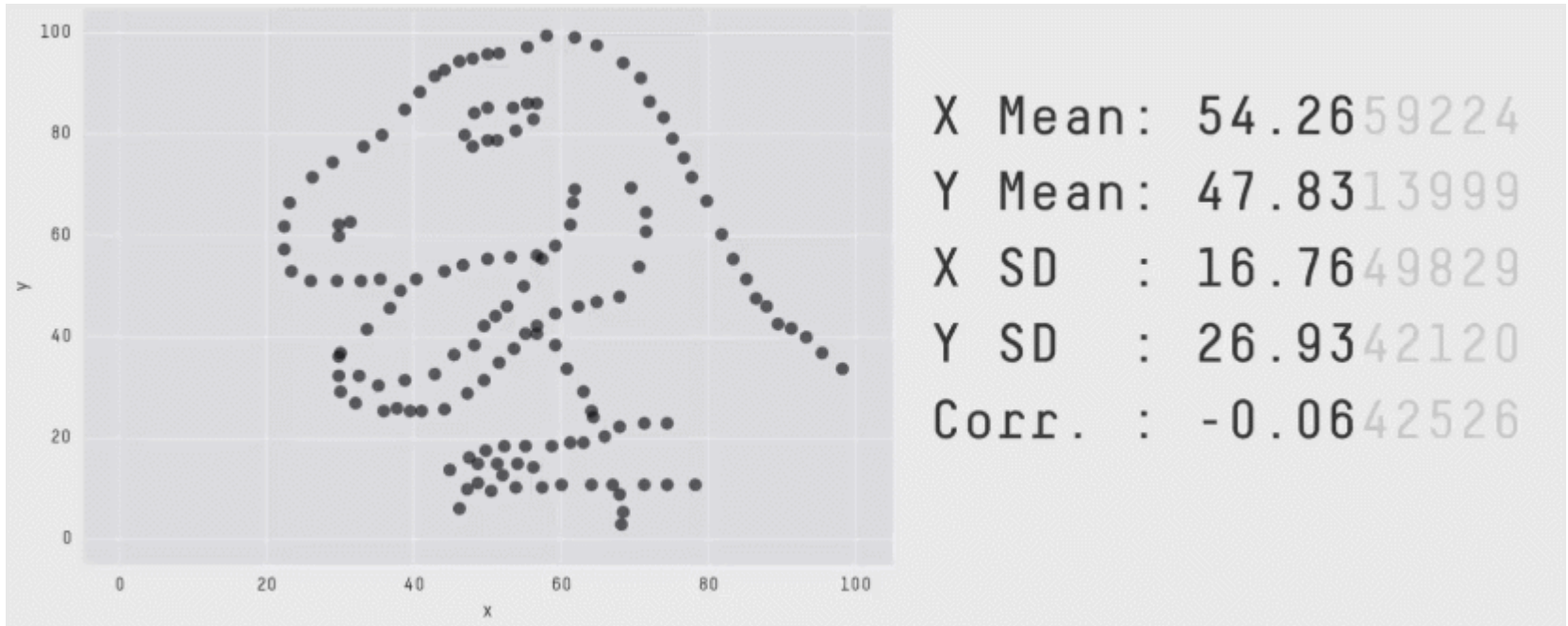
$$\text{Correl}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$



$$\text{Correl}(\text{Rainfall}, \text{Runoff}) = 0.91$$

Explain the data with Rep. Values?

❖ No! Proof?



Justin Matejka, George Fitzmaurice (2017). Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing CHI 2017 Conference proceedings: ACM SIGCHI Conference on Human Factors in Computing Systems

Summary (1)

❖ How to deal with Uncertainty?

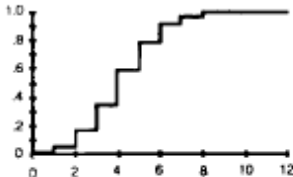
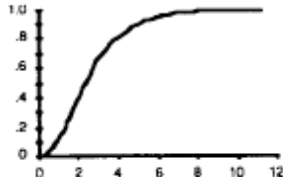
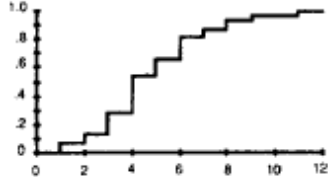
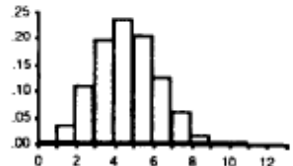
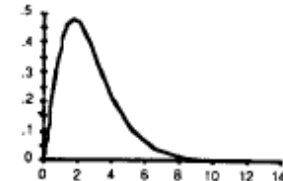
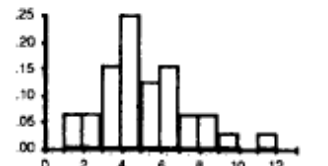
- ◆ Simplest approach = Replacing with representative values (대표값)
- ◆ Mean (expected values), median, worst-case values, etc. ...
- ◆ However, can create a poor representation of both the average performance & the possible performance range

❖ Common Representative Values for:

- 1) Measures of Central Tendency: mean, median, mode
- 2) Measures of Dispersion: range, IQR
- 3) Measure of Asymmetry: coefficient of skew
- 4) Measure of Correlation: linear correlation coefficient

Summary (1)

TABLE 17.2.1

Concept	Population value, discrete case	Population value, continuous case	Sample value
Cumulative distribution function (cdf)	 Describes the probability that a random variable is less than or equal to a specified value x	 Describes the probability that a random variable is less than or equal to a specified value x	 Empirical distribution function (edf): describes the observed frequency of a random variable being less than or equal to a specified value x
Probability mass function (pmf) and probability density function (pdf)	 pmf: the probability that X is equal to k	 pdf: first derivative of the cumulative distribution function $f(x) = \frac{dF(x)}{dx}$ $\mu = \int_{-\infty}^{\infty} xf(x) dx$	 Histogram: observed frequency with which random variable X falls into the assigned ranges $\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$
Mean, average, or expected value	$\mu = \sum_{i=1}^{\infty} P(X = x_i)x_i$		

Summary (1)

Variance	$\sigma^2 = \sum_{i=1}^n P(X = x_i)(x_i - \mu)^2$	$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$	$S^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$
kth central moment	$M_k = \sum_{i=1}^n P(X = x_i)(x_i - \mu)^k$	$M_k = \int_{-\infty}^{\infty} (x - \mu)^k f(x) dx$	$\hat{M}_k = \sum_{i=1}^n \frac{(X_i - \bar{X})^k}{n}$
Standard deviation		$\sigma = \sqrt{\sigma^2}$	$S = \sqrt{S^2}$
Coefficient of variation or relative standard deviation (if $\mu \neq 0$)		$CV = \frac{\sigma}{\mu}$	$CV = \frac{S}{\bar{X}}$
Coefficient of skew (a measure of asymmetry)		$\gamma = \frac{M_3}{\sigma^3}$	$G = \frac{\hat{M}_3}{S^3}$
Quantiles	x_p is any value of X that has the properties that $P(X < x_p) \leq p$ $P(X > x_p) \leq 1 - p$		$\hat{X}_p$ is the p th quantile of EDF
Median (useful for describing central tendency regardless of skewness)	$x_{0.5}$ Any value of X that has the property that $P[X < x_p] \leq 0.5$ $P[X > x_p] \leq 0.5$		$\hat{X}_{0.5}$ The middle observation in a sorted sample, or the average of the two middle observations if the sample size is even.
Upper quartile, lower quartile, and hinges	Upper quartile = $x_{0.75}$ Lower quartile = $x_{0.25}$		Upper hinge = $\hat{X}_{0.75}$ This is an approximation to the sample upper quartile; it is defined as the median of all sample values of $X \geq x_{0.50}$. The lower hinge, $\hat{X}_{0.25}$, is defined analogously.
Interquartile range (useful for describing spread of data regardless of symmetry)	$x_{0.75} - x_{0.25}$ Width of central region of population containing probability of 0.5		$\hat{X}_{0.75} - \hat{X}_{0.25}$ Width of central region of data set encompassing approximately half the data

Conclusion of Lecture 01

❖ Introduction to the Course: Statistical Hydrology

◆ Hydrology & Statistics

◆ 수문학에서의 불확실성 분류 방법 (Handbook of Hydrology, 17.1.1)

- **Inherent randomness** of the driving variables (predominantly **precipitation**) and the randomness of the hydrologic system (including topographic, aquifer, and soil characteristics)
- **Sampling error** – the measurements that hydrologists have to work with are only a small sample from a large or potentially infinite population
- Uncertainty resulting from incorrect understanding of the processes involved (**System error**- errors in estimating or predicting system outputs from system inputs)

◆ Introduction to Common Representative Values

- Central Tendency, Dispersion, Asymmetry, Correlation

Future Schedule Preview

❖ 다음 주 강의 (2022-09-13 화요일 18:30-21:20)

◆ Lec 02: Probabilistic Modeling in Hydrology- Part I (Moments & Probabilistic Models)

- Moments: Expectation, Variance, Skewness, Kurtosis
- Probabilistic Modeling:
Normal, Exponential, Gamma, Binomial, Poisson, etc.

❖ 2주 후 강의 (2022-09-20 화요일 18:30-21:20)

◆ Lec 03: Probabilistic Modeling in Hydrology- Part II (MLE & Goodness of fit tests)

- Maximum Likelihood Estimation of Parameters & Examples
- Goodness of fit Approaches such as:
AIC, PP or QQ plots, Chi-squared test

Datacamp Assignments

The screenshot shows the DataCamp website with a dark blue header. The header includes the DataCamp logo, a 'WE'RE HIRING' badge, and navigation links for 'Explore', 'Pricing', and 'For Business'. On the right, there are search, 'Sign In', and 'Get Started' buttons. The main content area features the headline 'Build data skills online' and the subtext 'Data drives everything. Get the skills you need for the future of work.' Below this are two buttons: 'Start Learning For Free' and 'DataCamp For Business'. On the right side, a 'Create Your Free Account' form is displayed, offering sign-up via Google, LinkedIn, or Facebook, or by email. The form includes fields for 'Email Address' and 'Password', and a 'Start Learning For Free' button. A disclaimer at the bottom of the form states: 'By continuing, you accept our Terms of Use, our Privacy Policy and that your data is stored in the USA.' The footer of the website lists various technologies and tools: python, R, PostgreSQL, + a b l e a u, Power BI, Excel, SQL Server, git, Shell, Scala, spark, and SPREADSHEETS.

datacamp WE'RE HIRING Explore Pricing For Business

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python R PostgreSQL + a b l e a u Power BI Excel SQL Server git Shell Scala spark SPREADSHEETS

Datacamp Assignments

- ❖ 서울대 이메일 주소로 Invite Link 송부 예정 (수업 종료 후)
- ❖ 학기 중에 과제의 일환으로 Datacamp Assignment 수행
- ❖ 다음주 목요일 오후 6시 이전까지 아래 Course 진행
 - ◆ Skill Track: R Programming - Introduction to R 완료

GRADING

Participation 10%

Assignments 30% (15% Datacamp Assignments & 15% In-class Assignments)

Quizzes 20%

Test 20%

Mid-Term Project 10%

Final Project 10%

⇒ 위의 명시되지 않은 부분에 있어서는 서울대학교 학칙 및 규정에 따름.

⇒ 늦게 제출되는 과제/프로젝트의 경우에는

제출기한 이후 3일까지 하루에 -10% (최대 -30%), 4-7일까지 -50%, 7일 초과시 0%.

2022-2학기 통계수문학 공지방 (카카오톡)

- ❖ 채팅방 이름: 2022-2 통계수문학
- ❖ <https://open.kakao.com/o/gEnH71ye>
- ❖ 참여 코드 0205

- ❖ 과제나 수업관련 공지 등을 진행할 예정
- ❖ 공지에 대한 대답 (따로 명시하지 않으면) 하지 않아도 괜찮습니다
- ❖ 참여 시, 이름을 "학번 성명" 으로 설정:
예) "2017-21430 김기주"